ST. CLARET COLLEGE (AUTONOMOUS), BENGALURU
Department of Bachelor of Computer Applications (BCA)

Vacation Mini Project Report

RecipeHub

Recipe Discovery & Management Web Application

Find Delicious Recipes Instantly.

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Semester	**4th Semester**
Submission	**July 2026**

https://recipe-hub-inchara.netlify.app/

CERTIFICATE

This is to certify that the mini project entitled

RECIPEHUB — RECIPE DISCOVERY & MANAGEMENT WEB APPLICATION

has been successfully developed and submitted by **INCHARA C V** (Roll No. **24BCA081**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **INCHARA C V**, Roll No. **24BCA081**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **"RECIPEHUB — RECIPE DISCOVERY & MANAGEMENT WEB APPLICATION"** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references and external APIs used have been duly cited and acknowledged.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

INCHARA C V

Roll No: 24BCA081 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to TheMealDB (www.themealdb.com) for providing a free, comprehensive recipe API that forms the data backbone of RecipeHub. Without this open API, building a recipe discovery application with thousands of real-world recipes would not have been possible.

I am also grateful to the open-source developer community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which guided my understanding of REST API integration, asynchronous JavaScript, localStorage management, responsive CSS design, and modular application architecture.

I thank my family for their constant support and encouragement throughout the vacation period.

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ABSTRACT

Cooking and recipe discovery are universal activities, yet finding reliable, well-organised recipes online often means navigating cluttered, advertisement-heavy websites with slow load times and poor mobile experiences. Students and home cooks need a clean, fast, and feature-rich recipe discovery tool that integrates search, favourites, custom recipe creation, and browsing history in a single, accessible interface.

RecipeHub is a responsive, multi-page static web application that connects to TheMealDB API to provide access to thousands of real-world recipes. Users can search by name or ingredient, browse trending recipes and curated categories, save favourites, create and store custom recipes locally, view detailed recipe pages with ingredients and step-by-step instructions, watch embedded YouTube tutorials, and track their browsing history — all within a polished dark/light themed interface.

Key Features:

- Recipe search by name with live suggestions powered by TheMealDB search.php API.
- Ingredient-based search using filter.php with multi-ingredient client-side filtering.
- Trending recipes section combining API results and custom local recipes.
- Category browser with pill-based filters fetching recipes by category from TheMealDB.
- Random recipe discovery using the random.php endpoint.
- Recently Viewed history tracking persisted in localStorage.
- Favourites management: save, search, and remove favourite recipes via localStorage.
- Custom Recipe Builder: create recipes with name, category, image URL, prep time, structured ingredients, instructions, and cuisine — stored locally as rh_custom.
- Recipe Detail page: ingredients list, numbered instructions, YouTube video embed, related recipes.
- Dark/light mode toggle persisted to localStorage.
- Toast notifications and reveal animations for enhanced interactivity.
- Fully responsive and mobile-friendly across all pages.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, TheMealDB REST API, localStorage API. Deployed on Netlify.

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INTRODUCTION

1.1 Background

Food is a central part of everyday life, and the ability to discover, save, and prepare diverse recipes is a skill that millions of people actively seek to develop. The internet hosts an enormous volume of recipe content, but most recipe websites prioritise advertisement revenue over user experience, resulting in cluttered layouts, slow page loads, and navigation that buries the actual recipe content beneath lengthy narrative introductions.

Open recipe APIs such as TheMealDB provide structured, machine-readable access to thousands of real-world recipes from global cuisines. A well-designed front-end application built on top of such an API can deliver a superior recipe discovery experience that is faster, cleaner, and more feature-rich than most commercial recipe websites.

1.2 About the Project

RecipeHub is a responsive, multi-page static web application that uses TheMealDB API as its data source to provide search-driven recipe discovery, category browsing, favourites management, custom recipe creation, and detailed recipe pages with embedded video tutorials. It is deployed as a static site on Netlify and requires no server-side infrastructure.

The application spans five HTML pages and is powered by a shared JavaScript module (js/app.js) that provides common utilities, API wrappers, localStorage managers, and UI components used across all pages.

Live at: **https://recipe-hub-inchara.netlify.app/** — deployed on Netlify.

1.3 Application Architecture

RecipeHub follows a multi-page static application architecture. A shared js/app.js module exports utility functions, API wrappers, and localStorage managers consumed by each page’s own script. Key shared managers include:

- **Favorites:** Manages rh_favorites in localStorage — add, remove, and check favourite status.
- **RecentlyViewed:** Manages rh_recent, storing recipe metadata for the home page history strip.
- **CustomRecipes:** Manages rh_custom, storing user-created recipes with structured ingredient rows.
- **Stats:** Manages rh_stats, tracking view counts per recipe for analytics.

- **apiFetch():** Centralised async fetch wrapper with standardised error handling for all TheMealDB API calls.

1.4 Scope

- Recipe discovery via TheMealDB API: search by name, ingredient, and category.
- Trending recipe browsing combining API and locally created recipes.
- Full recipe detail pages with ingredients, numbered instructions, and YouTube embeds.
- Favourites management with search and individual/bulk removal.
- Custom recipe creation and local storage management.
- Recently Viewed history and random recipe discovery.
- Dark/light mode, toast notifications, and reveal animations.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of RecipeHub:

1. To build a real-world API-integrated recipe discovery platform: Integrate TheMealDB's REST API endpoints to provide live recipe search by name, ingredient, category, and random discovery, giving users access to thousands of real-world recipes from global cuisines.

2. To implement multi-modal recipe search: Support three distinct search modes — by recipe name, by ingredient list (with multi-ingredient client-side filtering), and by cuisine category — making recipe discovery accessible for diverse user intentions.

3. To deliver a full recipe detail experience: Present each recipe's complete data including a structured ingredient-measure list, numbered step-by-step cooking instructions, embedded YouTube video tutorials, cuisine tags, and related recipe recommendations.

4. To enable personalised recipe management: Allow users to save favourite recipes to localStorage, search and remove saved favourites, and maintain a browsing history of recently viewed recipes with persistent state across sessions.

5. To provide a custom recipe creation tool: Build a structured recipe creation form that saves user-defined recipes to localStorage in a format compatible with the TheMealDB data structure, allowing custom recipes to appear alongside API results in search and trending sections.

6. To deliver a polished, responsive multi-page UI: Design a mobile-friendly dark-themed interface with toast notifications, scroll reveal animations, lazy image loading, and a dark/light mode toggle that maintains theme preference across page navigation.

7. To demonstrate modular front-end application architecture: Structure the application with a shared utility module (js/app.js) providing reusable functions across all five pages, demonstrating clean separation of concerns and scalable JavaScript architecture.

EXISTING SYSTEM

3.1 Current Recipe Discovery Tools

Users currently rely on a variety of platforms to discover and save recipes, each with notable limitations:

a) Commercial Recipe Websites (AllRecipes, BBC Good Food, Food Network)

These sites contain large recipe libraries but are heavily monetised with advertisements, auto-playing videos, and mandatory lengthy introductions before the recipe content. Page load times are slow and the mobile experience is often poor.

b) YouTube Recipe Videos

Video tutorials are popular but require watching the entire video to extract ingredients and steps. There is no structured data, no ingredient list export, and no integration with a searchable recipe database.

c) Recipe Apps (Yummly, Tasty, BigOven)

Mobile recipe apps require account creation, push notifications access, and often require subscriptions for offline or advanced features. They are not accessible as browser-based tools on all devices.

d) Bookmarking and Note Apps

Many users save recipe URLs in browser bookmarks or copy recipes into note apps. This approach is disorganised, offers no search or categorisation, and does not integrate with a structured recipe database.

3.2 Limitations Summary

- Commercial recipe sites are slow, cluttered, and advertisement-heavy.
- No free, clean, browser-based tool combines API search, favourites, and custom recipes.
- Ingredient-based search is rarely supported on consumer-facing recipe platforms.
- Custom recipe creation and storage is unavailable in most free tools.
- Most platforms require account creation and are not privacy-respecting.

PROPOSED SYSTEM

4.1 Overview

RecipeHub proposes a clean, fast, multi-page recipe discovery application that integrates TheMealDB's open API for live recipe data, augments it with locally created custom recipes, and provides a complete personal recipe management suite — all within a polished, responsive, dark-themed static web application.

4.2 Advantages

- **Clean, Ad-Free Interface:** No advertisements, no auto-play videos, no lengthy introductions. Recipe content is presented immediately and clearly.
- **Multi-Modal Search:** Search by name, by ingredient (with multi-ingredient client-side filtering), or by cuisine category — three modes in one interface.
- **Custom Recipe Integration:** User-created recipes are stored locally and appear alongside API results in search and trending sections, using the same card format.
- **Full Detail Pages:** Ingredients, numbered steps, embedded YouTube tutorials, cuisine tags, and related recipes on a single, comprehensive detail page.
- **No Account Required:** Favourites, history, custom recipes, and theme preference are all persisted in localStorage with no login needed.
- **Responsive and Mobile-Friendly:** Designed to work across screen sizes with mobile navigation, touch-friendly cards, and optimised layouts.

4.3 API Integration Architecture

RecipeHub uses four TheMealDB API endpoints via a centralised apiFetch() wrapper:

Endpoint	URL Pattern	Used For
search.php?s=	Search by recipe name	Name-based search, trending
filter.php?i=	Filter by ingredient	Ingredient-based search
filter.php?c=	Filter by category	Category browsing
random.php	Random meal	Random recipe discovery
lookup.php?i=	Lookup by ID	Recipe detail page
categories.php	All categories	Category pill navigation

MODULES AND FEATURES

RecipeHub comprises five HTML pages, each serving a distinct user purpose.

5.1 Home Page (index.html)

The home page is the primary discovery interface, featuring a full-width hero section with a food photography background overlay.

Hero Section:

A large bold headline — "Find *Delicious* Recipes Instantly" — with a name/ingredient search bar, quick-search category pills (Pasta, Chicken, Sushi, Pizza, Curry, Beef), and three live stats counters showing 300+ Recipes, Saved count, and Viewed count from localStorage.

Trending / Popular Recipes:

A loadTrending() function fetches a mixed batch of API recipes and custom local recipes, rendered as recipe cards with category badge, heart favourite button, recipe name, cuisine, and a View Recipe button.

Recently Viewed Strip:

A horizontal scrollable strip of thumbnail cards for all recipes in rh_recent localStorage, allowing quick return to previously explored recipes.

Browse All Categories:

A grid of category cards fetched from TheMealDB categories.php endpoint, each showing a food image and category name. Clicking a category loads all matching recipes via filter.php?c=.

Ingredient Search Section:

A separate ingredient-based search input that fetches by primary ingredient and optionally filters results client-side for multi-ingredient queries.

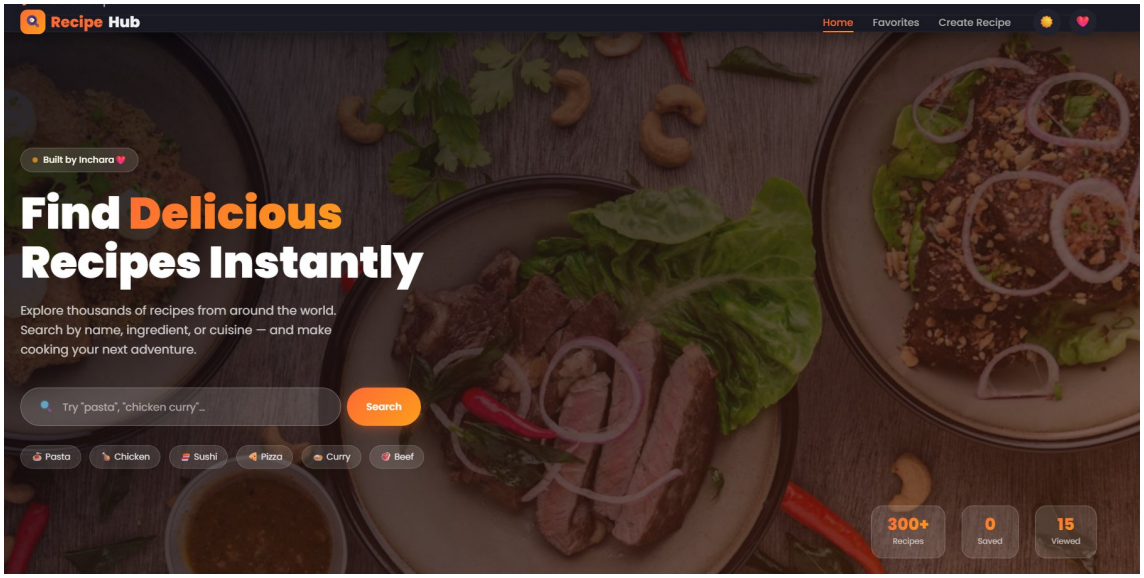


Figure 5.1 – Home Page Hero Section with Search, Category Pills, and Stats

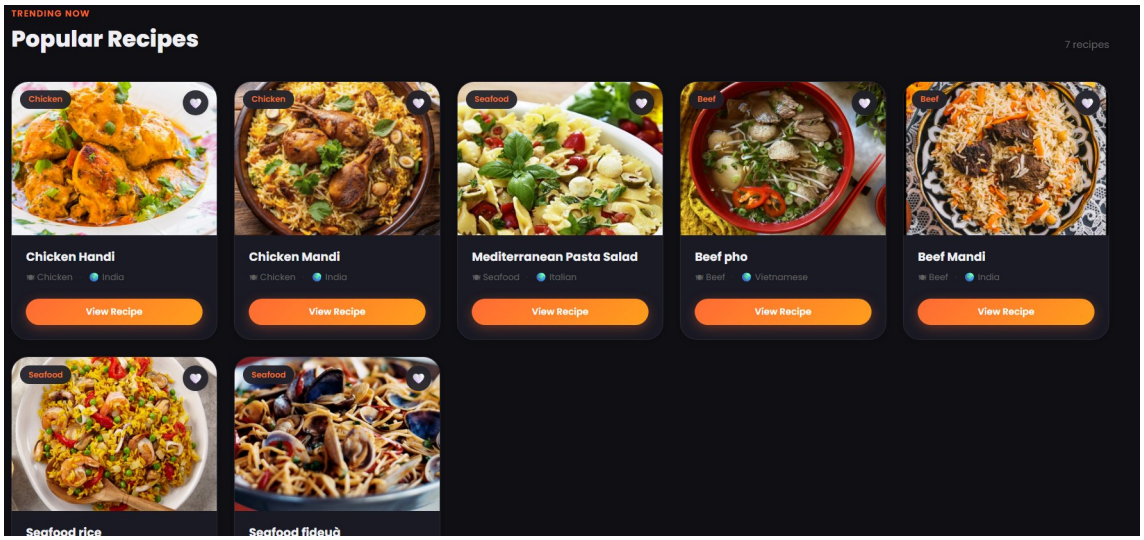


Figure 5.2 – Trending / Popular Recipes Section with Recipe Cards

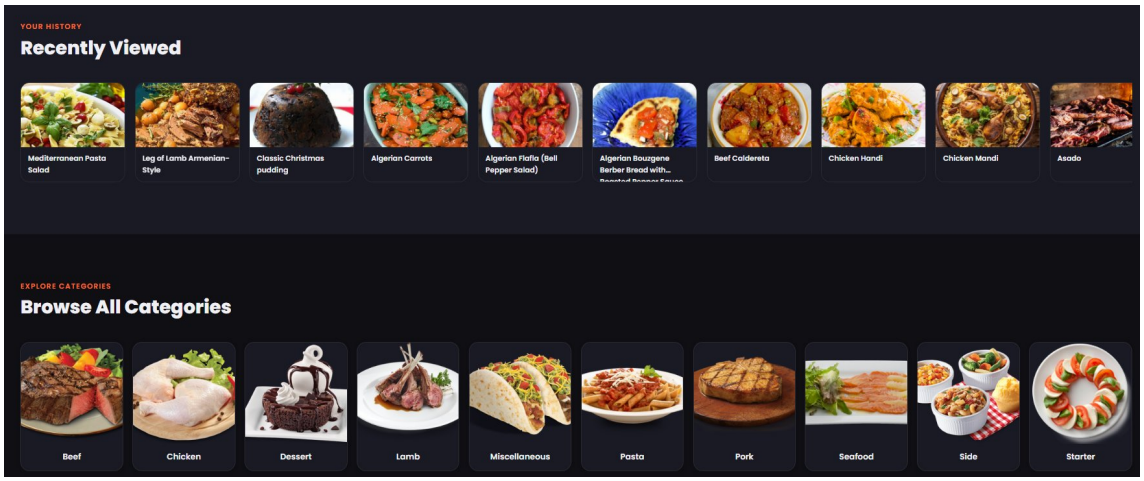


Figure 5.3 – Recently Viewed History Strip and Browse All Categories Grid

5.2 Recipe Detail Page (recipe.html)

The recipe detail page provides the complete recipe experience, loaded dynamically from the URL query string ID parameter using lookup.php?i=.

Ingredients Panel:

A structured list of all ingredients with their measures, rendered from the TheMealDB's strIngredientN / strMeasureN field pairs into a clean ingredient-quantity table format.

Instructions Panel:

The recipe's instruction text is split by line breaks and rendered as individually numbered steps with orange numbered circle indicators, making the cooking process easy to follow sequentially.

Video Tutorial Embed:

When a YouTube URL is available in the recipe data (strYoutube field), an embedded YouTube iframe is displayed in a Watch Video Tutorial panel, allowing users to watch the cooking demonstration without leaving the page.

Related Recipes:

A More Like This section fetches additional recipes from the same category and displays them as cards below the main recipe content.

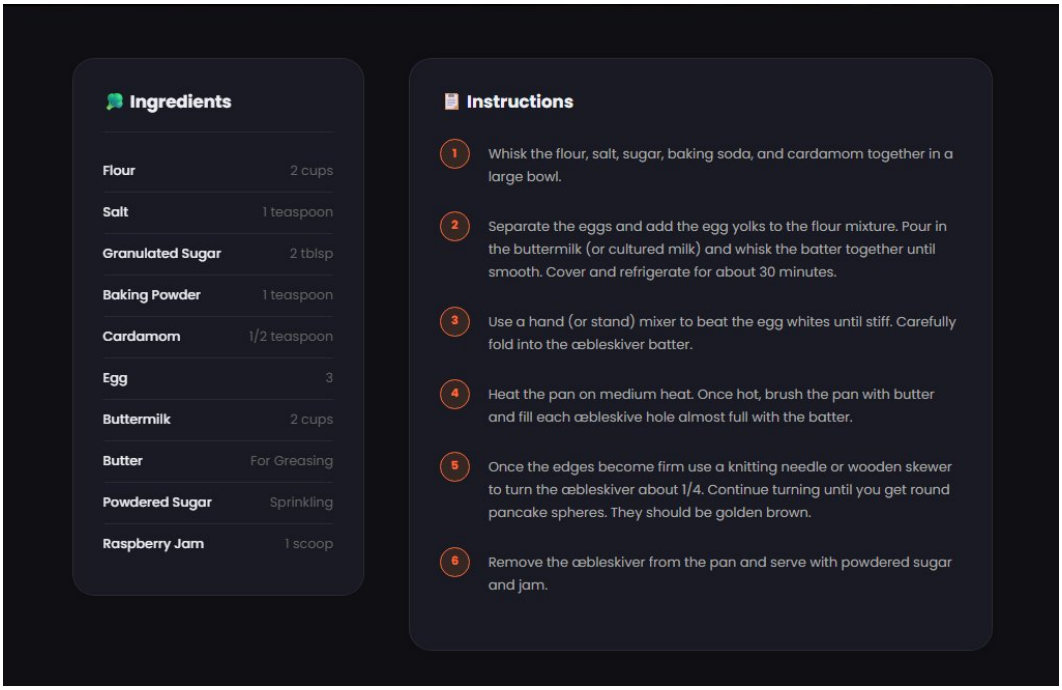


Figure 5.4 – Recipe Detail Page: Ingredients and Numbered Instructions

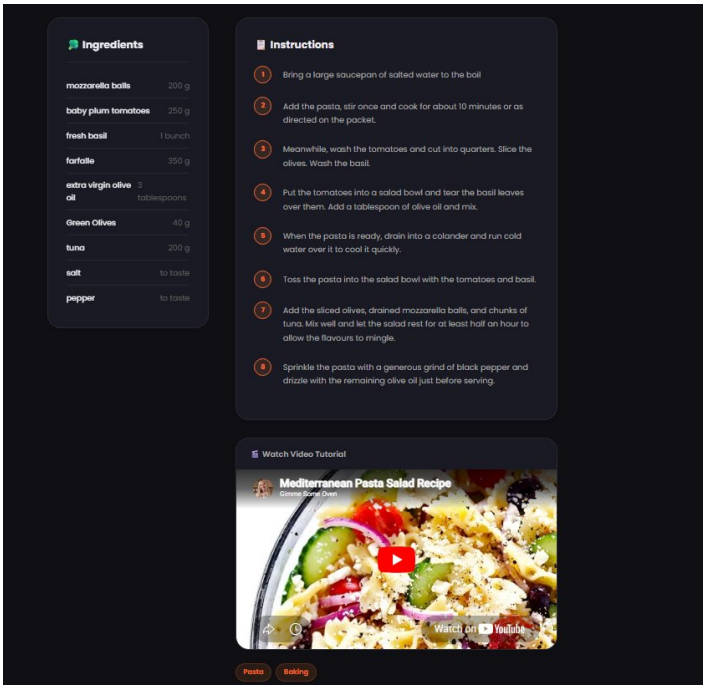


Figure 5.5 – Recipe Detail Page: Instructions, YouTube Video Embed, and Tags

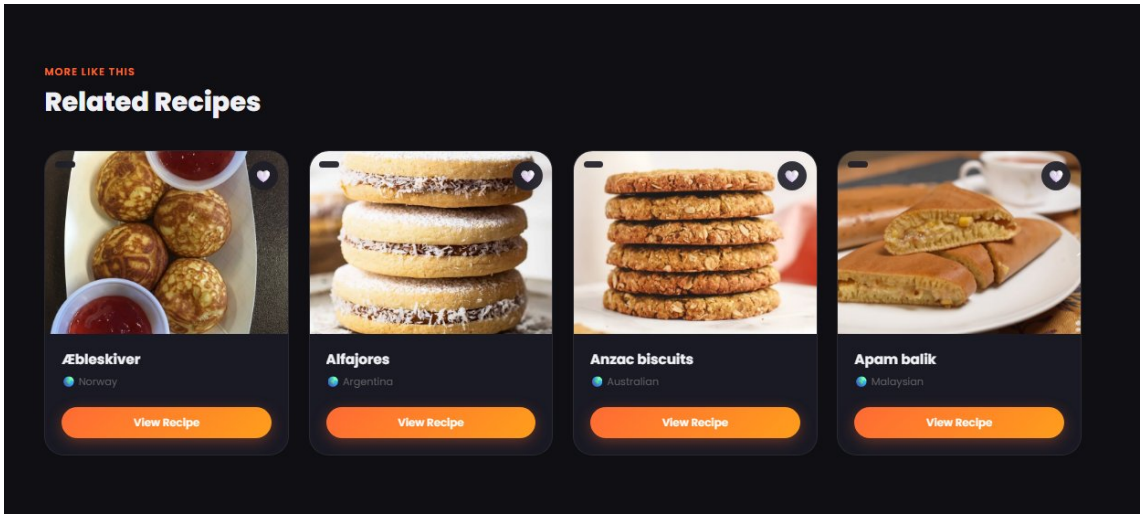


Figure 5.6 – Related Recipes Section on the Recipe Detail Page

5.3 Favourites Page (favorites.html)

The Favourites page displays all recipes saved to rh_favorites in localStorage.

A search bar filters saved favourites by name in real time. Each favourite card shows the recipe image with category badge and filled heart icon, the recipe name, cuisine, a View Recipe button linking to the detail page, and a Remove button that deletes the recipe from localStorage and updates the displayed list. A Clear All option removes all saved favourites at once.

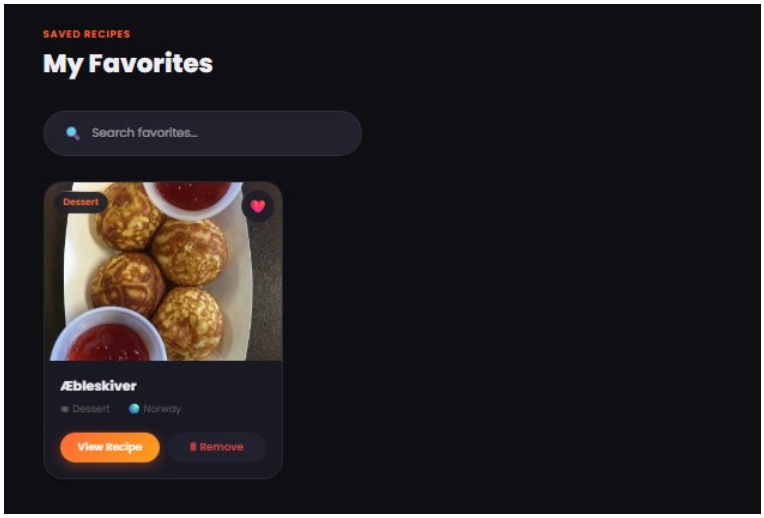


Figure 5.7 – Favourites Page with Saved Recipe Card

5.4 Create Recipe Page (create.html)

The Create Recipe page provides a structured form for building and saving custom recipes to rh_custom in localStorage.

Form Fields:

- **Recipe Name** (required): The display name of the custom recipe.
- **Category** (required): Dropdown matching TheMealDB categories for compatibility.
- **Prep Time (minutes)** (optional): Preparation time in minutes.
- **Image URL** (optional): URL of an image to display on the recipe card.
- **Ingredients** (required): Structured rows of ingredient name and amount pairs, with + Add Ingredient for additional rows.
- **Instructions** (required): Free-text area where each new line becomes a numbered step on the detail page.
- **Cuisine / Origin** (optional): The recipe's regional cuisine or origin.

On saving, ingredients are serialised into both structured rows and strIngredientN/strMeasureN fields for full compatibility with the shared recipe card rendering system. Custom recipes appear alongside API results in trending and search sections.

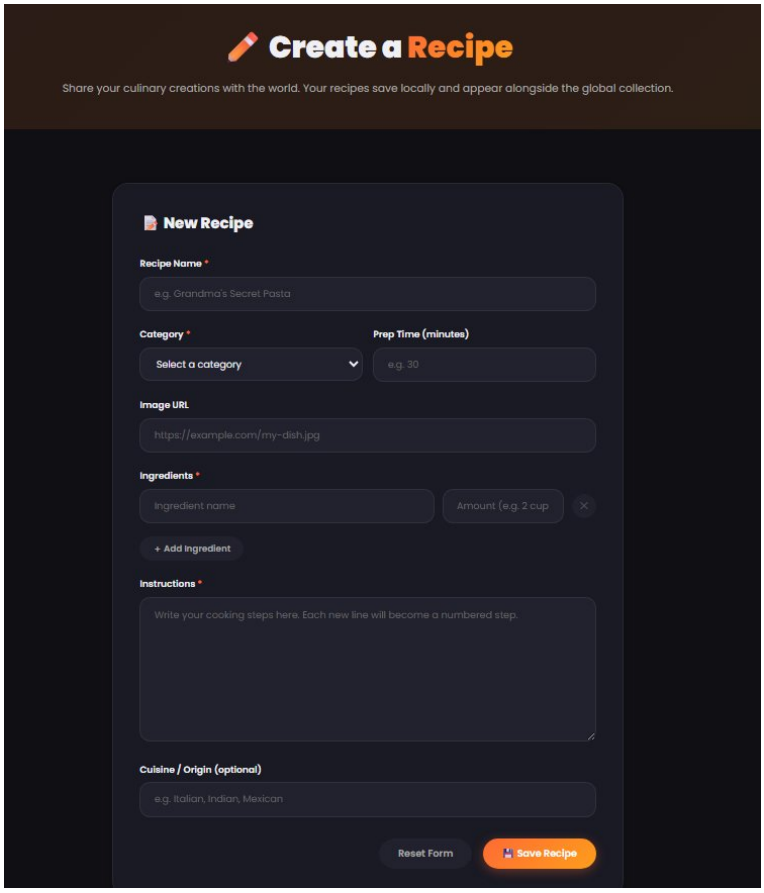


Figure 5.8 – Create Recipe Page with New Recipe Form

TECHNOLOGIES USED

RecipeHub was developed using native front-end web technologies and one external API. No build tools, frameworks, or back-end server are used.

6.1 HTML5

HTML5 provides the structural foundation for all five pages of RecipeHub. Semantic elements organise the navigation, hero sections, recipe card grids, and detail page layouts. The YouTube embed uses an HTML5 iframe element. Form controls on the Create Recipe page use standard HTML5 input types. Data attributes on recipe cards and category pills store IDs and metadata for JavaScript event handling.

6.2 CSS3

CSS3 delivers RecipeHub's polished dark-themed interface with orange accents. CSS Custom Properties manage both the dark and light theme colour systems, switching via a class toggle on the root element and persisted to localStorage. CSS Grid layouts the recipe card grid and category browser. Flexbox handles the navbar, hero section, and ingredient-measure rows. CSS transitions and keyframe animations power the scroll reveal effects. The application is fully responsive via CSS media queries across mobile, tablet, and desktop breakpoints.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript. The shared js/app.js module provides: the centralised apiFetch() async wrapper for all TheMealDB API calls, the Favorites, RecentlyViewed, CustomRecipes, and Stats localStorage manager classes, shared utility functions for debounce, toast notifications, dark mode, navbar scroll behaviour, scroll reveal animations, and lazy image loading, and the shared renderRecipeCard() function used consistently across all pages. Page-specific scripts handle the home page flows (loadTrending, loadCategoryPills, searchRecipes, ingredient search, random recipe), the recipe detail page (ingredient extraction from strIngredientN fields, step parsing, video embed), the favourites page (load, search, remove), and the create page (form handling, ingredient row management, localStorage save).

6.4 TheMealDB REST API

TheMealDB (www.themealdb.com/api/json/v1/1/) is a free, open recipe API providing access to thousands of meals from global cuisines. RecipeHub uses six endpoints: search.php?s= for name search, filter.php?i= for ingredient search, filter.php?c= for category filtering, random.php for random discovery, lookup.php?i= for full recipe detail by ID, and categories.php for the full category list. All

calls are made via the centralised apiFetch() wrapper which standardises error handling and response parsing.

6.5 localStorage API

The Web Storage API's localStorage interface persists all user-specific state across sessions using five keys: rh_favorites (saved recipe list), rh_recent (recently viewed recipe metadata), rh_custom (user-created recipes), rh_stats (view count analytics), and rh_theme (dark/light mode preference). Custom recipes are stored in a TheMealDB-compatible format so they integrate seamlessly with the shared card rendering system.

6.6 Netlify

RecipeHub is deployed as a static multi-page site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution. As a static site with no server-side code, Netlify's free tier is fully sufficient.

#	Technology	Type	Purpose
1	HTML5	Markup	5-page structure, forms, iframes, data attributes
2	CSS3	Styling	Dark/light themes, responsive grid, animations
3	JavaScript ES6+	Programming	API calls, CRUD, shared utilities, rendering
4	TheMealDB API	External REST API	Recipe data source for search and details
5	localStorage	Browser API	Favourites, history, custom recipes, theme
6	Netlify	Platform	Static multi-page hosting and CDN

CONCLUSION

RecipeHub is a comprehensive, well-architected recipe discovery and management application that successfully integrates a live REST API with a rich set of local storage features to deliver a complete culinary companion for home cooks and food enthusiasts. By combining TheMealDB's extensive recipe database with locally managed favourites, browsing history, and custom recipe creation, the project delivers a genuinely useful tool that addresses real limitations of existing recipe platforms.

The technical breadth of RecipeHub is one of its most notable qualities. Integrating six distinct API endpoints, building a shared JavaScript module architecture across five HTML pages, implementing three search modes (name, ingredient, category), managing four localStorage keys with type-safe manager classes, and delivering a fully responsive dark/light themed UI — all in Vanilla JavaScript without any framework — represents a substantial front-end development achievement.

The decision to make custom user-created recipes compatible with the TheMealDB data structure — serialising them into strIngredientN/strMeasureN fields and integrating them seamlessly with API results in search and trending sections — demonstrates careful data architecture thinking that goes beyond typical student project scope.

On a personal level, building RecipeHub during the vacation period has been an immensely valuable experience in working with real-world external APIs, designing modular JavaScript architectures, and creating a polished multi-page application that functions as a genuine, deployable product.

In conclusion, RecipeHub is a polished, feature-rich, and publicly deployed recipe discovery application that fully achieves its stated objectives and stands as a strong demonstration of API integration skills, front-end architecture, and independent learning initiative.

Live Project: https://recipe-hub-inchara.netlify.app/